Nikhil R. Agrawal - Curriculum Vitae

Process Engineer (H1B Visa holder)
Deposition Product Group, Lam Research
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EDUCATION

Ph.D. in Chemical Engineering August 2024
University of California, Berkeley, USA
Specialization area: Statistical physics, Electrical Double Layers, Interfacial Science, Chemical Physics
Thesis title: *Modified Gaussian Renormalized Fluctuation Theory for Electrolytes at Interfaces*
Thesis Committee: Rui Wang (Advisor), Clayton Radke, David Limmer, and Kranthi Mandadapu

B. Tech. and M. Tech (Dual Degree) in Chemical Engineering July 2018
Indian Institute of Technology (IIT), Delhi, India
Institute Silver Medal for highest GPA in the program
Specialization area: Multiphase Reactors, Techno-economic analysis of processes, Mathematical Modeling
Master's thesis title: *Microscopic characterization of particle packings in packed bed reactors*
Master's thesis advisor: Prof. Shantanu Roy

WORK EXPERIENCE

Designing PEALD processes and chambers for thin film materials *Sep '24 - Present*
Process Engineer | PEALD Deposition Product Group, Lam Research, Oregon, USA

- Hands on experimental work and rigorous material characterization to develop next generation low k (<3.5) ALD Liner for Through Silicon Vias (*patent pending*).
- Hardware component evaluation to improve throughput for Lam's Striker Chambers
- Evaluated process performance of a new RF system for Lam's next generation of Striker Chambers
- Screened multiple ALD precursors and reagents to developed carbon and fluorine doped ALD Oxide and Nitride films, gaining experience in crucial organic chemical reactions in semiconductor processing
- Coordinated with and drove a multinational team of software, mechanical, electrical, and RF plasma engineers, together with Lam's Field Service Engineers in Taiwan, to ensure smooth project execution
- Developed a PECVD SIN diffusion barrier film, and solved the defects problem for oxide-nitride stack in Lam's Striker chambers

Statistical physics of electrical double layers in concentrated electrolytes *Aug '18 - Aug '24*
Ph.D. Candidate | Chemical Engineering, University of California, Berkeley, USA

Original research work to explain how highly concentrated salt solutions behave near charged surfaces, leading to 7 first author research publications and multiple conference presentations. Key contributions include:

- Developed a new theoretical model to accurately model short and long range electrostatic correlation in electrolytes
- Explained counterintuitive phenomena in systems with multivalent salts, including charge inversion, like-charge attraction, and opposite-charge repulsion
- Developed a new spectral method based algorithm to efficiently calculate two point electrostatic correlation operator

Advanced ML and Bayesian Modeling for quantifying uncertainty in material models *Mar '24 - June '24*
Semester Intern | Physics Division, Lawrence Livermore National Laboratory, USA

Collaborated with a 10+ member multidisciplinary team to deploy Bayesian and deep learning methods for quantifying uncertainty in materials simulations, developing practical experience in AI/ML methods applicable across domains.

Data Science, Cloud Computing, and Monte-Carlo simulations for complex systems *June '23 - Aug '23*
Summer Intern | Center of Computational Biology, Flatiron Institute, New York, USA

Conducted large-scale hybrid Monte-Carlo and Molecular dynamics simulations to model protein behavior during cell division, building capabilities in Monte-Carlo methods, cloud-based computation, and scientific data engineering.

Reactor Catalyst-Packing optimization for efficiency gains in refining operations *Jan '17 - July '18*
Master's Candidate | Department of Chemical Engineering, IIT Delhi, India

Designed and validated a mathematical model describing catalyst-bed microstructure, providing insights that could improve packing-methods and led to 2 publications.

Large scale computer simulations to optimize catalyst packing in packed bed reactors *May '17 - July '17*
Visiting Researcher | Mentor: Prof. Dr. Thorsten Pöschel, University of Erlangen-Nuremberg, Germany

Image processing and pore-network modeling of nuclear reactors *May '16 - July'16*
Summer Intern | Porous Media Group, Institut de Mecanique des Fluides de Toulouse (IMFT), France

Design of solar-powered photocatalysts for conversion of CO_2 into commodity chemicals *May'15 - April '16*
Undergraduate Researcher | Mentor: Prof. Suddhasatwa Basu, Chemical Engineering, IIT Delhi, India

Photo-catalysis is the use of semiconducting materials and solar energy-generated electrons to drive chemical reactions, a principle shared with photovoltaic devices. In this project, I

- synthesized and characterized structure of semiconductor–graphene/silica nanocomposite catalysts, and
- analyzed the photo-catalytic activity of these catalysts through dye degradation in a *self-designed reactor*.

JOURNAL PUBLICATIONS

On the topic of electrolysis, electrolytes, and polyelectrolytes:

1. Electrostatic Correlation Augmented Self-Consistent Field Theory and Its Application to Polyelectrolyte Brushes
   Chao Duan, Nikhil R. Agrawal, and Rui Wang, *Physical Review Letters (2025) 134, 048101*
2. Understanding long-range opposite-charge repulsion in multivalent salt solutions
   Nikhil R. Agrawal, Carlo Carraro and Rui Wang, *J. Chem. Phys. 161, 204902 (2024)*
3. Nature of overcharging and charge inversion in electrical double layers
   Nikhil R. Agrawal, Chao Duan, and Rui Wang, *J. Phys. Chem. B 2024, 128, 1, 303–311*
4. Ion correlation-driven like-charge attraction in multivalent salt solutions
   Nikhil R. Agrawal, Ravtej Kaur, Carlo Carraro and Rui Wang, *J. Chem. Phys. 159, 244905 (2023)*
5. Non-monotonic salt concentration dependence of inverted electrokinetic flow
   Nikhil R. Agrawal and Rui Wang, *AIChE Journal, e18269, 2023.*
6. Self-consistent description of vapor-liquid interface in ionic fluids
   Nikhil R. Agrawal and Rui Wang, *Physical Review Letters (2022), 129, 228001.*
7. Electrostatic correlation induced ion condensation and charge inversion in multivalent electrolytes
   Nikhil R. Agrawal and Rui Wang, *Journal of Chemical Theory and Computation (2022), 18, 6271-6280*

8. Sturm–Liouville theory inspired method to solve the Modified Gaussian Renormalized Fluctuation theory for electrolytes
   Nikhil R. Agrawal, Carlo Carraro and Rui Wang, *in preparation.*

On the topic of trickle bed reactors, multiphase reactors, and process intensification:

1. A first-order segregation phenomenon in fluid-immersed granular systems
   Prapanch Nair, LAT Cisneros, CRK Windows-Yule, Nikhil R. Agrawal, Shantanu Roy, and Thorsten Pöschel, *Powder Technology 373 (2020): 357-361.*

2. Isotropy of sphere packings in a cylindrical confinement
   Nikhil R. Agrawal, Prapanch Nair, Thorsten Pöschel and Shantanu Roy, *Chemical Engineering Journal 377 (2019): 119820.*

AWARDS AND HONORS

- Langmuir Graduate Student Oral Presentation Awards Finalist, American Chemical Society 2022
- *Institute Silver Medal* from IIT Delhi for highest GPA in the dual degree program in Chemical Engg. 2018
- IIT Delhi *Semester Merit* Award for 9 out of 10 semesters for being among *top 7%* meritorious students across the dual degree program, *consecutively for 8 semesters* 2013-2017
- *Significant Contribution to Research Activities* Award by Chemical Engineering Society, IIT Delhi 2017
- *Kishore Vaigyanik Protsahan Yojana* (KVPY) Fellowship by Dept. of Science and Technology, India 2013
- Was among the *Nation's Top 1%* merit holders in National Standard Examination in Physics (NSEP) conducted by the Indian Association of Physics Teachers (IAPT) 2013

LEADERSHIP ROLES

Secretary, SPIC MACAY, BRCA, IIT Delhi *2015 - 2016*
SPIC MACAY: *Society for Promotion of Indian Classical Music And Culture Amongst Youth*

One of the 11 club secretaries of the Board of Recreational and Cultural Activities (BRCA), IIT Delhi.

- Led a team of 13, one each from every hostel in IIT Delhi, to organize marketing, publicity, and hospitality of club events like dance workshops and music concerts
- Successfully secured a sponsorship of Rs. 1 lac from the Gas Authority of India Ltd. (GAIL) to organize *Saaz*, the classical music night of Rendezvous 2015 (IIT Delhi's annual cultural festival), where Grammy Award winner Pt. Vishwa Mohan Bhatt performed.
- Overall coordinator of SPIC MACAY's VIRASAT '16, annual classical music festival of IIT Delhi; headed a team of 15 to organize a 2 week long series of cultural events

Event Coordinator, IIT Delhi Alumni Association *April '16*
Lead a team of student volunteers for audience management in a seminar on "Ethics and Happiness" by *Nobel Laureate* 14th *Dalai Lama*

MISCELLANEOUS

Programming Skills	Python, C++, PyTorch, GPyTorch ,TensorFlow, Scikit Learn, Bash
Technical softwares	JMP, MATLAB, Aspen, COMSOL, FLUENT, ParaView, Autodesk Inventor
Material Characterization	XSEM, STEM, SIMS, XRR, FTIR, CV/IV (Hg Probe).
Other Interests	Sociology, Reading about emerging technologies and their impact on society